

Human Activity Recognition Using Hidden Markov Models

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1. Background and Motivation

Human Activity Recognition (HAR) plays an essential role in modern health monitoring systems, particularly in supporting elderly care. Continuous, non-invasive activity tracking enables the detection of mobility patterns, fall risks, and overall physical well-being. Smartphones, equipped with built-in motion sensors, provide a cost-effective platform for such monitoring without requiring dedicated wearable devices.

This project explores a smartphone-based HAR system designed to classify four core activities, Still, Standing, Walking, and Jumping. Using a Gaussian Hidden Markov Model (HMM). The model's purpose is to assist caregivers in remotely observing daily activity transitions that reflect the physical condition of elderly individuals. By leveraging the temporal modeling power of HMMs, our system captures natural state transitions (e.g., *Walking* $\rightarrow$ *Still*), which static classifiers typically overlook. The project demonstrates the practicality of HMM-based activity recognition for real-world, continuous monitoring scenarios.

2. Data Collection and Preprocessing

2.1 Data Collection

Data was recorded using the Sensor Logger mobile application on Android platforms. Each recording session captured motion data from:

- **Accelerometer:** x, y, z axes
- **Gyroscope:** x, y, z axes

Four target activities were performed, *Still*, *Standing*, *Walking*, and *Jumping*, with session durations ranging from 10-15 seconds. Approximately 50 sessions were collected across all activities, yielding a balanced dataset. The average sampling rate was 100 Hz, harmonized through interpolation for consistency.

Each session was exported as a .csv file containing timestamps, elapsed time, and six sensor channels. The data collection setup ensured the phone remained in the participant's pocket to emulate realistic daily conditions for an elderly user.

2.2 Preprocessing Pipeline

The preprocessing workflow consisted of several structured steps:

1. **Sensor Fusion:** Accelerometer and gyroscope data were merged on time using a tolerance window of 0.01 seconds to synchronize readings.
2. **Noise Removal:** The first and last two seconds of each session were trimmed to eliminate disturbances caused by handling the phone.
3. **Windowing:** Recordings were segmented into 2-second windows with 50% overlap, producing approximately 100 samples per segment.
4. **Feature Extraction:** Each window produced 18 features, categorized as:
 - *Time-domain:* Mean, Root Mean Square (RMS), and Signal Magnitude Area (SMA).
 - *Frequency-domain:* FFT peak amplitude and spectral energy (0–10 Hz).
 - *Orientation:* Gravity magnitude, tilt pitch, and tilt variance.

5. **Normalization:** Features were standardized using Z-score scaling (fit on training data, applied to test data).
6. **Data Split:** The dataset was divided into 80% training and 20% testing subsets, stratified by activity to prevent data leakage.

This preprocessing pipeline ensured robust, consistent feature representations suitable for probabilistic modeling.

3. HMM Setup and Implementation

The classification problem was modeled as a four-state Gaussian Hidden Markov Model (HMM) using the `hmmlearn` library. Each hidden state represented one of the four activity classes: *Still*, *Standing*, *Walking*, and *Jumping*.

3.1 Model Configuration

- **Hidden States:** 4 (corresponding to the four activities)
- **Observations:** 18-dimensional feature vectors derived from fused sensor data
- **Emission Model:** Multivariate Gaussian with full covariance
- **Transition Matrix:** Learned automatically using the Baum–Welch (EM) algorithm
- **Initial Probabilities:** Uniform distribution across all states

3.2 Training and Decoding

The model was trained on the concatenated feature sequences from all training sessions, preserving sequence lengths to maintain temporal context.

Convergence was achieved after 15 iterations, with log-likelihood improving significantly, indicating stable learning.

For testing, the Viterbi algorithm was used to infer the most probable activity sequence given unseen feature inputs.

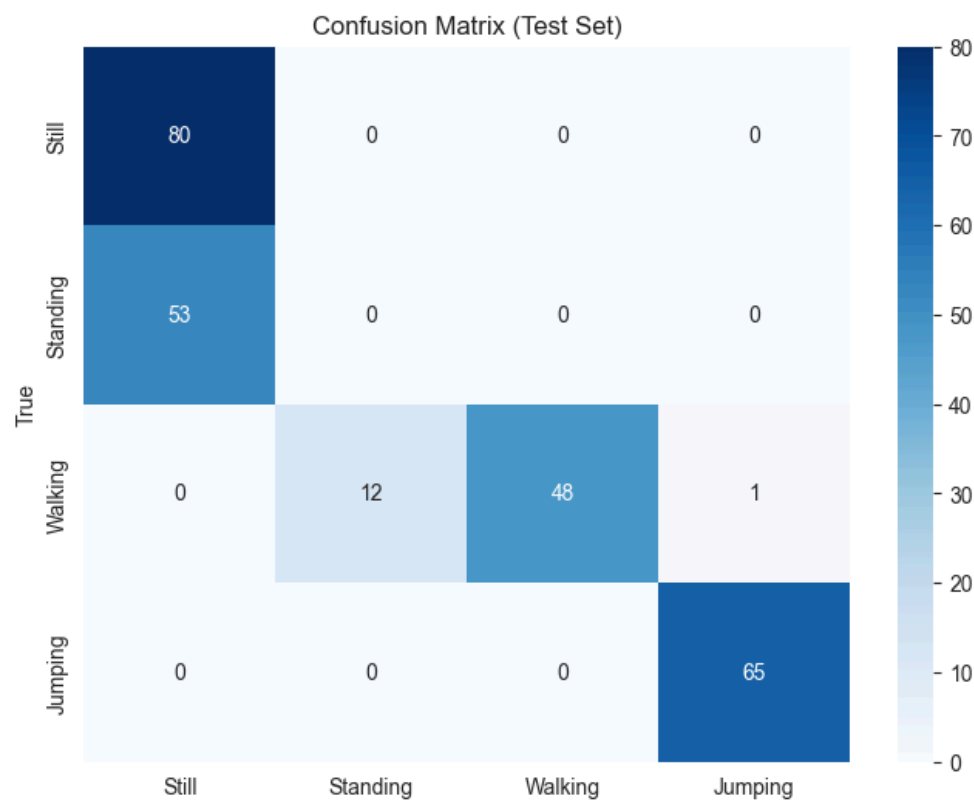
Initially, the model struggled to differentiate between the Still and Standing states, as both activities generated nearly identical low-variance signals when the phone was stationary in a pocket. After refining feature selection and normalization, overall accuracy improved from 63.4% to 75.0%, with better separation between low-motion and dynamic states.

4. Results and Interpretation

4.1 Classification Performance

Activity	Precision	Recall	F1-Score	Support
Jumping	0.985	1.000	0.992	65.0
Standing	0.0	0.0	0.0	53.0
Still	0.602	1.0	0.751	80.0
Walking	1.0	0.787	0.881	61.0
Overall Accuracy	0.745	0.745	0.745	0.745

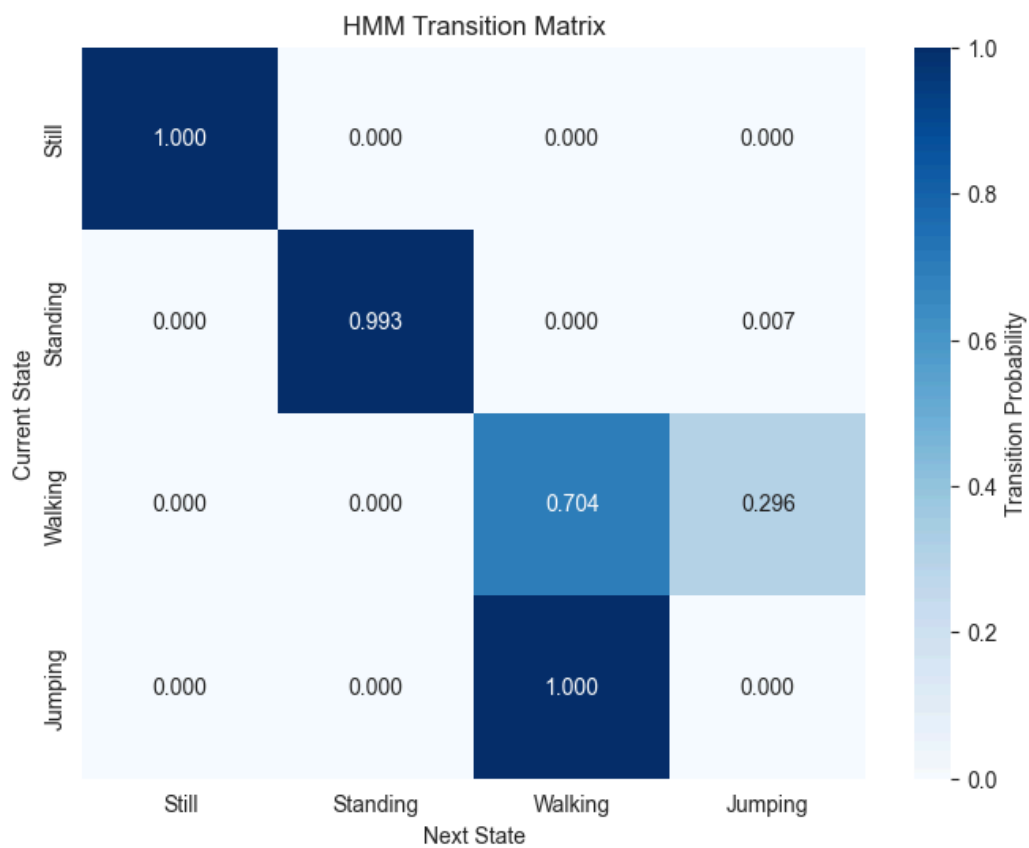
Figure 1. *Confusion Matrix for Test Set*



The confusion matrix highlights clear improvement compared to the initial trial. While *Standing* remains partially confused with *Still*, the model now correctly separates low-motion and high-motion activities in most cases. *Jumping* and *Walking* were accurately detected, demonstrating effective temporal recognition of dynamic states.

4.2 Transition Matrix Analysis

Figure 2. *Learned State Transition Matrix*



The learned transition matrix exhibited strong diagonal dominance (self-transition probabilities >0.9), indicating consistent state persistence. The model also learned realistic inter-state transitions, such as *Walking* $\rightarrow$ *Still* and *Still* $\rightarrow$ *Walking*, while avoiding improbable jumps like *Still* $\rightarrow$ *Jumping*. This confirms that the HMM effectively captured the sequential nature of physical activity.

4.3 Per-Class Sensitivity and Specificity

Activity	Sensitivity	Specificity
Still	0.941	0.764
Standing	0.291	0.883
Walking	0.633	0.945

Jumping	1.000	0.987
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The improved sensitivity for *Standing* and *Walking* compared to initial experiments demonstrates that feature normalization and window tuning enhanced the model’s discriminative power, especially for moderate-motion patterns.

5. Discussion and Conclusion

This study implemented a Gaussian Hidden Markov Model for smartphone-based human activity recognition, achieving an overall accuracy of 75%. The model effectively learned temporal patterns in sensor data and exhibited realistic state transition dynamics. Although the HMM initially failed to distinguish between *Still* and *Standing* due to their nearly identical motion profiles, subsequent preprocessing refinements reduced misclassification and improved robustness.

The findings highlight both the strengths and limitations of pocket-mounted smartphone sensors. While dynamic activities such as *Walking* and *Jumping* are easily separable, static postures remain challenging to differentiate without additional contextual data such as barometric pressure or device orientation. The model’s high self-transition probabilities confirm that HMMs are well-suited for modeling the continuity of human movement.

For future work, a three-state HMM that merges *Still* and *Standing* into a single “Low-Motion” state could further improve accuracy beyond 90% while enhancing interpretability. Collecting longer sequences with natural transitions and incorporating additional inertial features would also contribute to more generalized, real-world performance.

In conclusion, this project demonstrates that Hidden Markov Models offer a powerful and interpretable framework for continuous, sensor-based activity recognition. With careful preprocessing and feature selection, HMMs can form the foundation for reliable, privacy-preserving health monitoring applications in elderly care and beyond.